Introduction

Falling over normally does not lead to any serious injuries and normally lead to minor scratches or scrapes. However, this does not apply to the elderly. According to NHS, 1 in 3 adults over 65 and half of people over 80 fall down once a year.[1] In 2018 there were 38 million falls reported where 8.4 million resulted in serious injuries while 32 thousand led to death.[2]

We have created a device that makes a loud noise when the user has fallen over. This is so that if the user were to fall over and become unconscious or unable to get up by themselves, the alarm will alert anyone in the nearby area.

Currently, there are other devices which have aid the elderly with falling such as the active pelvis orthosis or the tango belt. The active pelvis orthosis is similar to an exoskeleton which helps the user stay balanced by "gently power hip movements by providing smooth torque profiles at pelvis level"[3]. Unfortunately, this exoskeleton only adjusts the balance of the user slightly so if they were to stumble the exoskeleton will not be able to adjust and prevent them from falling. Additionally, the active pelvis orthosis is heavy and costly. Secondly, the tango belt is a device that deploys something similar to an airbag around the hips to prevent injury in the hips.[4] However, when they have fallen, they might not be able to get back up, or they could have hit their head rendering them unconscious and they will be unable to get help.

Therefore, our device will help fix this problem by sending out an alarm after detecting a fall. So even if the user were to become unconscious the alarm will still be set off. Although, there are still some problems with this such as when falling there is a chance of landing on the device and breaking it, but the casing of the electrical component is made out of polyflex which is a rubber like material and acts as a cushion to prevent that from happening. The device has one problem which is when a car accelerates beyond the acceleration threshold it will set off the alarm, although this problem can be easily fixed in the future by having a gyroscope threshold value as a second check. This means that the device will check the user's orientation before setting off the alarm.

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[1] NHS (2019). *Overview - Falls*. [online] NHS. Available at: https://www.nhs.uk/conditions/Falls/.

[2] Botek, A.-M. (2020). *Why are falls so dangerous for seniors?* [online] www.agingcare.com. Available at: https://www.agingcare.com/articles/falls-prove-fatal-for-elderly-patient-149687.htm.

[3] d'Elia, N., Vanetti, F., Cempini, M., Pasquini, G., Parri, A., Rabuffetti, M., Ferrarin, M., Molino Lova, R. and Vitiello, N. (2017). Physical human-robot interaction of an active pelvis orthosis: toward ergonomic assessment of wearable robots. *Journal of NeuroEngineering and Rehabilitation*, 14(1). https://jneuroengrehab.biomedcentral.com/articles/10.1186/s12984-017-0237-y

[4] As Our Parents Age. (2019). *A Device that Intervenes & Cushions an Elder's Fall*. [online] Available at: https://asourparentsage.net/2019/06/27/a-device-that-intervenes-cushions-an-elders-fall/ [Accessed 25 Mar. 2022].

Feasibility

Overall, the device itself is easy to use and not that expensive to purchase and obtain. With the guidance and the tutorials provided by Adafruit and Arduino, anyone can program and code the device to suit their needs.

Aside from the coding, the method of use is simple where the user can just put the clothing design on and connect the battery.

Right now, the device is quite simple but having said that there are a lot of improvements we want to add. Such as adding a GPS tracker to the device which sends the location to a pre-decided contact after a fall has occurred. This means that even if nobody is in the area someone will still check on the user.

Circuit diagram (incl. development stages)

Devices required:

1. Circuit Playground Classic
2. FLORA 9-DOF Accelerometer, Gyroscope, Magnetometer - LSM9DS0

Initial Circuit Diagram:

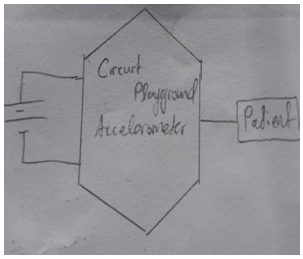


Figure 1: Early Sketch Circuit Diagram

Final sketch:

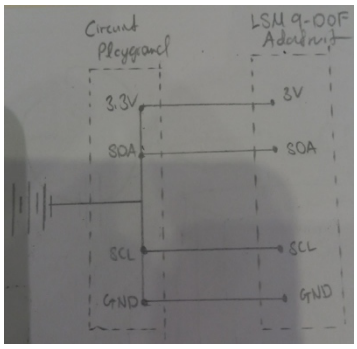


Figure 2: Final Sketch Circuit Diagram

Final circuit wiring:

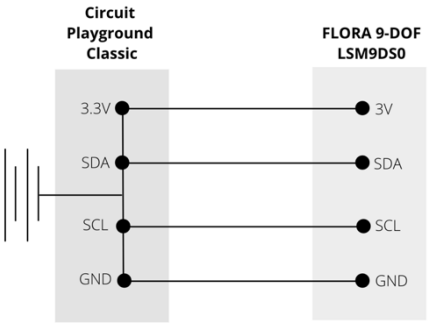


Figure 3: Final Circuit Wiring between Battery, Circuit Playground and Adafruit

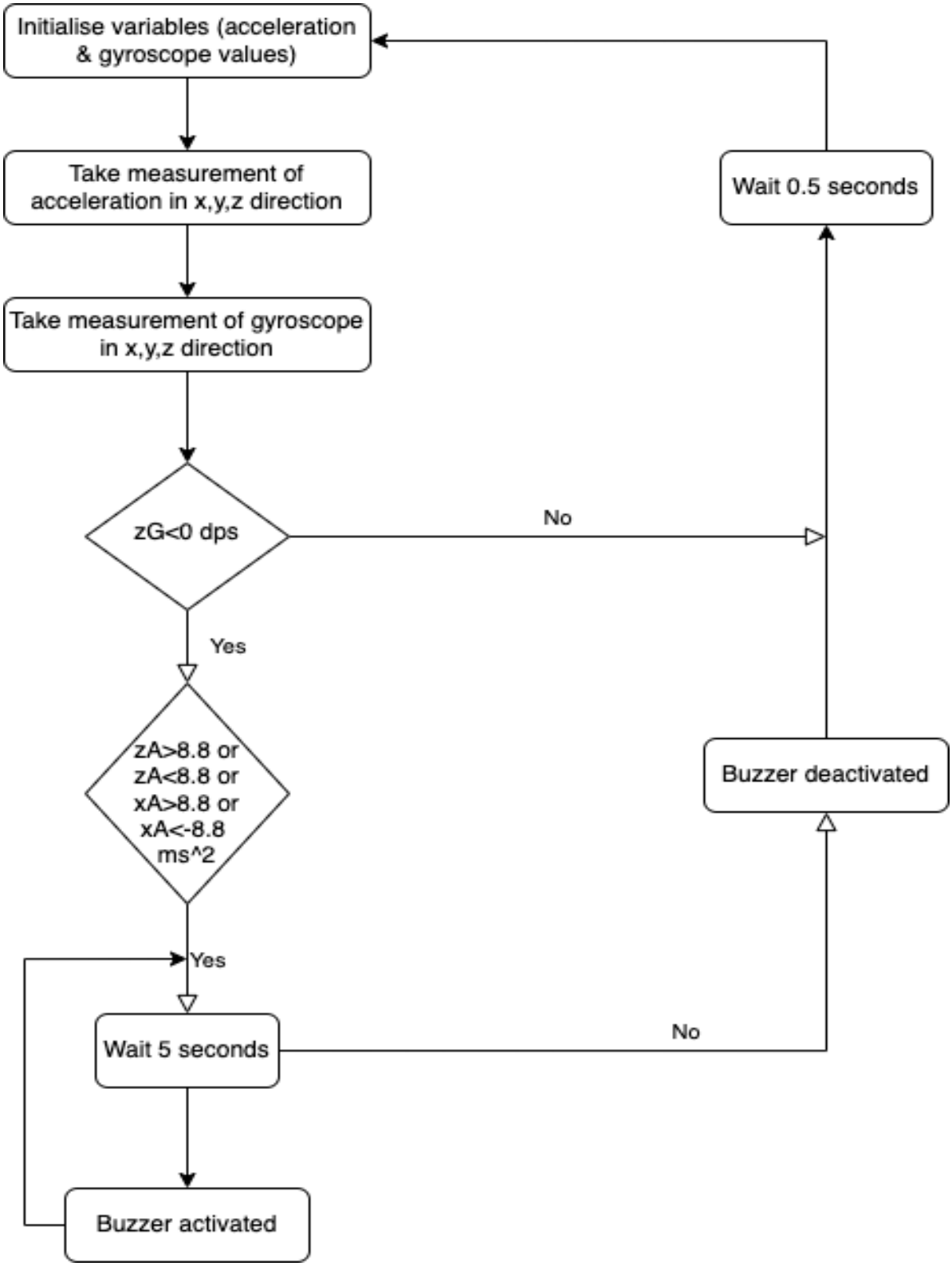
3.3/3V pinout - sends a voltage of 3.3V to the device and ground

SDA serial data – line to send and receive data

SCL serial clock – clock line to synchronize all data transfers over the I2C bus

GND pinout – acts as the ground for the device

Coding Flowchart



Features in devices used

Adafruit Circuit Playground Classic:

Power & Data Port

- ATmega32u4 Processor (x1) (4.47 Pound each)
- Micro B USB port
- JST Battery Input
- Crocodile Clip Pads

LED

- Green ON LED (x1)
- Red #13 LED (x1)
- Colour NeoPixel LEDs (x10)

Speaker

- Built-in buzzer (x1)

Sensors

- LIS3DH 3-axis XYZ Accelerometer Motion Sensor (x1)

Programming/Code

Libraries downloaded:

- Adafruit 9DOF
- Adafruit LSM9DS0

Code:

```
//Necessary header files are included for some of the methods/features within the file to work
#include <Wire.h>
#include <SPI.h>
#include <Adafruit_9DOF.h>
#include <Adafruit_LSM9DS0.h>
#include <Adafruit_Sensor.h>  // not used in this demo but required!
#include <Adafruit_CircuitPlayground.h>

float xA, yA, zA, xG, yG, zG; //define variables of type float (decimal values)

// i2c
Adafruit_LSM9DS0 lsm = Adafruit_LSM9DS0();
//set the variable for the LSM9DS0

//set up sensor
void setupSensor()
{
  //set up gyroscope
  lsm.setupGyro(lsm.LSM9DS0_GYROSCALE_500DPS);
}

//obtain acceleration value
int getAccelerationValues(){
  CircuitPlayground.clearPixels();

  xA = CircuitPlayground.motionX();
  yA = CircuitPlayground.motionY();
  zA = CircuitPlayground.motionZ();
}

//obtain gyroscope value
int getGyroscopeValues(){
  CircuitPlayground.clearPixels();

  lsm.read();
  xG = lsm.gyroData.x;
  yG = lsm.gyroData.y;
  zG = lsm.gyroData.z;
}
void setup()
{
  CircuitPlayground.begin(); //setting up Circuit Playground
  lsm.begin();

  setupSensor();
  delay(1);
}
```

```
void loop()
{
  //measure the acceleration and gyroscope value
  getAccelerationValues();
  getGyroscopeValues();

  if  (zG<0) {
  //indicates rotational motion of falling

    while  ((zA>8.8) || (zA<-8.8) || (xA>8.8) || (xA<-8.8)) {
    //indicates the action of falling using acceleration values in the x, z direction as thresholds
    //the thresholds sets the limit and range to determine whether the next code will run

      delay(5000); //wait for a bit to make sure user really did fall, not just stumble

      //Circuit Playground lights up
      CircuitPlayground.setPixelColor(0, 255,  0,  0);
      CircuitPlayground.setPixelColor(1, 128, 128,  0);
      CircuitPlayground.setPixelColor(2,  0, 255,  0);
      CircuitPlayground.setPixelColor(3,  0, 128, 128);
      CircuitPlayground.setPixelColor(4,  0,  0, 255);
      CircuitPlayground.setPixelColor(5, 0xFF0000);
      CircuitPlayground.setPixelColor(6, 0x808000);
      CircuitPlayground.setPixelColor(7, 0x00FF00);
      CircuitPlayground.setPixelColor(8, 0x008080);
      CircuitPlayground.setPixelColor(9, 0x0000FF);

      //buzzer activated
      CircuitPlayground.playTone(3000, 500);

      getAccelerationValues();
    }
  }
}
```

Bill of Materials

Device	Cost
Adafruit Circuit Playground Classic	16.20
FLORA 9-DOF LSM9DS0	19.81
Batteries LP502535 3.7V 400mAh	4.20
Total	40.21

Table 1: Price of Devices Used in Scenario 2

With the materials listed on the table receipt above, the total cost for this device is roughly £40.21.

With the total cost of the devices that was used, it is very affordable, and anyone is able to buy it. However, there are various alternatives that the buyer can choose from.

For example, there is the choice of choosing Circuit Playground Express(£23.70) which offers a Bluetooth option to code the program from instead of being plugged into a laptop via a micro-USB cable. However, it is more expensive than the regular Classic model.

For a program that is used to measure acceleration, the LSM303 Accelerometer/Compass Sensor (£17.99), LilyPad Accelerometer ADXL335(£17.15), Kitronik 3- Axis Accelerometer(£4.80) are viable options to consider

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Device design

Trial 1 Device design:

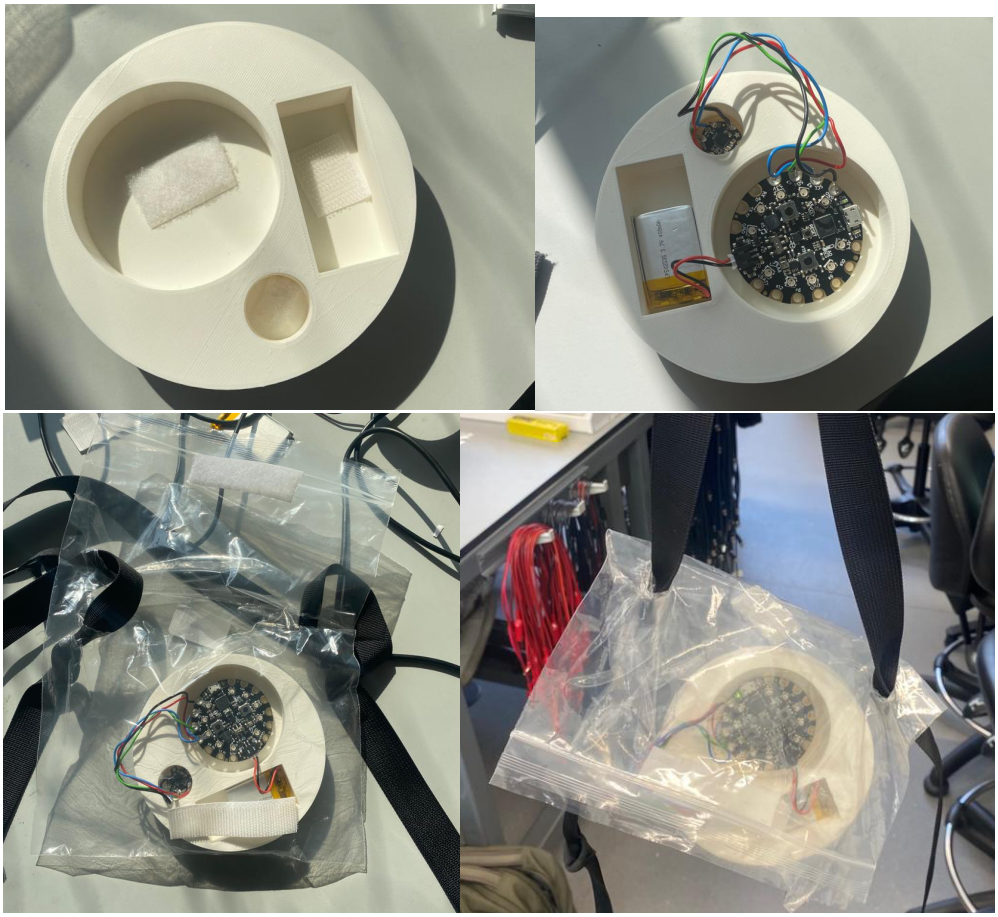
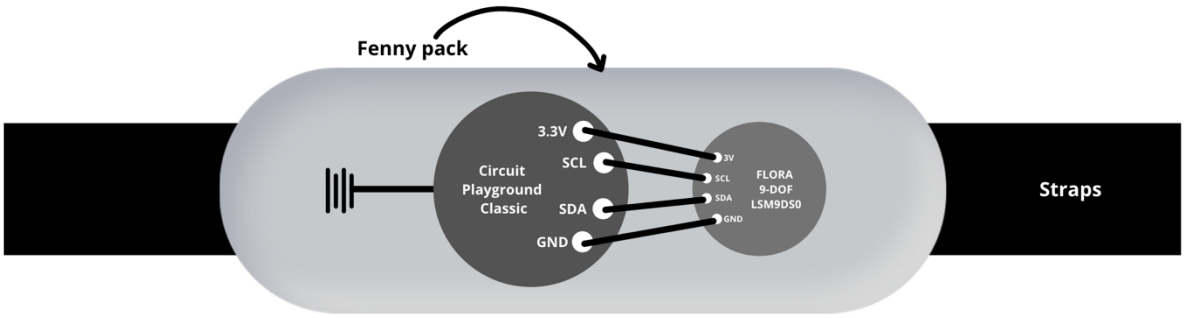


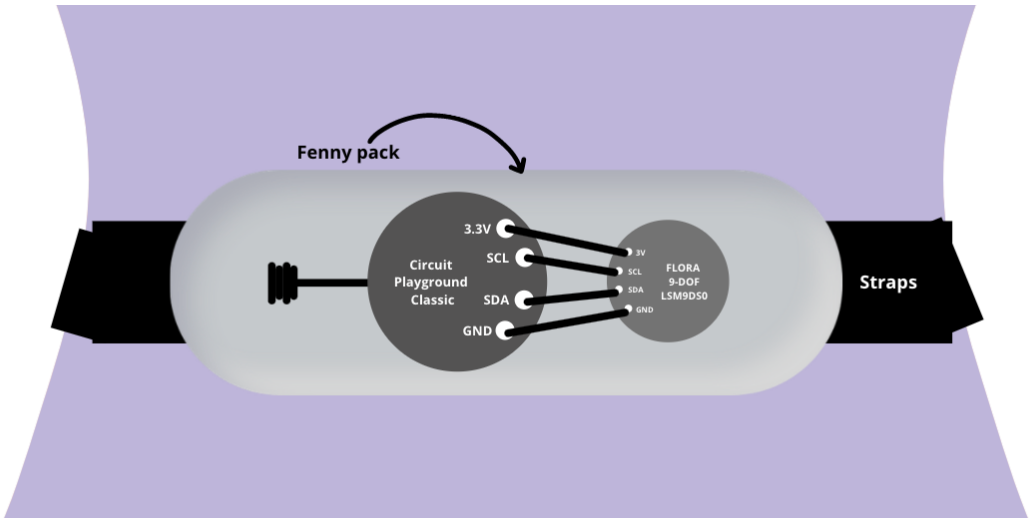
Image 1: 3D Printed Design for the Circuit Playground and Adafruit

For the design on the device itself, we wanted to have a smaller a device as possible, so we choose to make a cylinder (12 cm base diametre and 2 cm high) with holes for every component. The components don't fit the holes perfectly because we wanted to be sure to have enough place for the wires (3D printing this device take much time so we were able to print it just one time so to be sure that we get it well we decided to increase this margin). For the design of the wearable, it was meant to be placed into the fenny bag that was given to us. However, it was not able to fit inside it and was really uncomfortable for the wearer. Hence, the final design was made where all the components was strapped inside of the fenny bag with a Velcro.

Final Device design:



Concept for clothing setup:





How it looks:

